Homework Academic Integrity & AI Use Pledge

Course: 可信賴之人工智慧

Homework / Assignment: HW1

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Student Pledge

I affirm that I have completed this assignment with academic integrity and that the submitted work represents my own understanding, judgment, and effort. I understand that **the use of generative AI tools, including ChatGPT, Claude, Gemini, and similar systems, is permitted in this course unless otherwise specified by the instructor or the assignment instructions.** However, AI should be used as a learning and productivity tool, not as a substitute for my own learning.

I therefore pledge that:

1. **I will not simply submit AI-generated content as my own work.**
   When I use AI to generate answers, explanations, code, designs, reports, or other content, I will review, revise, and integrate the material based on my own understanding.

2. **I will take responsibility for all submitted work, including AI-assisted work.**
   I understand that using AI does not transfer responsibility for the submitted work to the AI tool. I am responsible for its correctness, reasoning, quality, and integrity.

3. **I will make a genuine effort to understand the AI-generated content that I use.**
   This includes, when applicable, understanding the **code, algorithms, system architecture, mathematical reasoning, explanations, designs, or other content** generated or suggested by AI.

4. **I will independently verify AI-generated content before submitting it.**
   I will check for errors, unsupported claims, incorrect calculations, inappropriate assumptions, security or implementation issues, and other potential problems.

5. **I will modify and adapt AI-generated content when necessary.**
   I understand that meaningful use of AI may require correcting, rewriting, simplifying, extending, or otherwise adapting its output so that the final submission reflects my own understanding and meets the requirements of the assignment.

6. **I will be able to explain the work that I submit.**
   If requested, I will provide a written explanation, demonstration, presentation, or recorded explanation showing that I understand the submitted work and can explain the relevant **code, system, methodology, reasoning, or other AI-assisted content**.

7. **I will disclose my use of AI when required by the instructor or assignment.**
   When requested, I will identify the AI tool(s) used and describe how AI contributed to my work, including relevant prompts, outputs, or modifications when appropriate.

8. **I will use AI to support learning rather than circumvent it.**
   The purpose of homework is to develop my own knowledge, reasoning, problem-solving, and technical skills. I understand that simply obtaining an answer from AI without understanding the underlying work does not fulfill this purpose.

I understand that **AI-assisted work may be considered my own academic work only when I have appropriately reviewed, understood, verified, and, when appropriate, modified the AI-generated content in accordance with the requirements of this course.**

By signing below, I confirm that I have read, understood, and agreed to this pledge. I also acknowledge that I may be asked to demonstrate my understanding of any AI-assisted work that I submit.

Student Signature: 藍璟誠

Date: 115.10.3

Homework 1: Train Your Own Small GPT and Evaluate It on TMMLU+

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1. Setup and evaluation protocol

Model: MiniMind (commit f659b55), dense, hidden size 768, 8 layers, vocabulary 6,400, 63.91M parameters. Checkpoints are converted with MiniMind's own conversion function to the standard Hugging Face Qwen3ForCausalLM format (strict weight loading).

Hardware: one NVIDIA GeForce RTX 4070 Ti SUPER (16 GB), WSL2 Ubuntu 24.04, PyTorch 2.14.0 (CUDA 13.0), transformers 4.57.6.

Evaluation: eval_tmmluplus.py with its defaults (0-shot, no chat template, log-likelihood over the letters A–D); the official metric is MACRO accuracy over the four categories.

Data protocol: The TMMLU+ v1.1 validation split (2,197 questions, 66 subjects) is split per subject into a 20% held-out set (466 questions, seed 42) and an 80% part (1,731) that may be used for training. Because the private grading set reorders answer options, we also evaluate an option-shuffled copy of the held-out set (same questions, permuted options). All development decisions use only the held-out set; on the practice set (2,001 test questions) we evaluated only the baseline and the final model, plus, after the final choice was fixed, the model produced by our reproduction run (the author's minimind-3 checkpoint was also evaluated once as a sanity check of the pipeline). After the first four Stage-3 experiments (E1–E4) and before running E5–E7, we fixed a decision rule: a change counts as an improvement only if the mean micro accuracy over the two held-out versions exceeds the baseline (25.32%) by at least 2 points, the tie threshold used in the assignment's leaderboard grading.

2. Baseline (Task 1)

The baseline uses MiniMind's default configuration: pretraining (2 epochs, max_seq_len 340) followed by SFT (2 epochs, max_seq_len 768) on the mini datasets, seed 42. The pretraining loss falls from 7.5 to about 1.9 and the SFT loss from 2.1 to about 1.5 (Appendix B). **The baseline scores 21.91% MACRO on the practice set (micro 23.59 ± 0.95)** and 22.49% MACRO on the held-out set, i.e. below random guessing.

Diagnosis: the baseline does not read the question. Its held-out predictions are 54.5% A, 35.4% D and under 6% each for B and C (Fig. 2). A model that ignored the question but had this letter preference would be expected to score $\Sigma_l P(\text{pred} = l) \cdot P(\text{gold} = l) = 23.6\%$ on the original and 26.1% on the shuffled held-out set; the observed scores are 24.46% and 26.18%. The baseline's accuracy is therefore fully explained by letter bias, and its below-chance practice score comes from that bias meeting the practice set's answer distribution. The author's released minimind-3 checkpoint, evaluated the same way, also scores 24.73% MACRO on the practice set, so this reflects the model scale and data rather than a pipeline error.

3. What we changed and why (Task 2)

3.1 Tokenizer (direction 2)

We measured how the MiniMind tokenizer handles the evaluation prompts, comparing each validation prompt with its OpenCC-simplified version (Table 1). **The same content costs 40.9% more tokens in Traditional script**, and **38.6% of Chinese character occurrences are split into several byte-level tokens**, versus 8.2% in Simplified. The most frequently split characters are everyday Traditional forms (為, 於, 時, 關, 與, 會, 國, 學), whose Simplified forms are single tokens. Sequence length, by contrast, is not a bottleneck: only 5.3% of prompts exceed the 340-token pretraining length.

Validation prompts (2,197)	Traditional (original)	Simplified (OpenCC)
Mean tokens per prompt	194.7	138.2
Characters per token	0.84	1.19
Chinese character occurrences split into >1 token	38.6%	8.2%
Prompts longer than 340 / 768 tokens	5.3% / 1.1%	2.2% / 0.3%

Table 1. MiniMind tokenizer on TMMLU+ validation prompts. A–D remain single tokens in both cases.

Change: we added a character-level Traditional→Simplified normalizer (3,553 mappings generated from OpenCC alone, no TMMLU+ data) to tokenizer.json. This needs no retraining, reduces prompts to 138.2 tokens on average, keeps A–D as single tokens and costs about 2 ms per question.

3.2 Multiple-choice format alignment (direction 3)

Because the baseline never saw the answer format, we fine-tuned it on TMMLU+ train (317 questions) plus the 80% validation part (1,731), after removing 2 questions identical to held-out items (2,046 questions in total). Three design choices: (a) prompts are built with the evaluation script's own build_prompt, and the scores computed before training match the official baseline exactly, which confirms that format and scoring are aligned; (b) the loss is computed only on the answer letter; (c) **option rotation:** every question appears four times with the correct option placed at A, B, C and D, so the

labels are perfectly balanced and the letter itself carries no information. The control without rotation repeats each question four times in its original order, so both settings take the same number of steps.

3.3 Hyperparameters (direction 4) and evaluation options

For the Stage-3 fine-tuning we varied the learning rate (2e-5, 1e-4, 3e-4) and the number of epochs (2, 3, 10); batch size 16, AdamW, 5% warm-up with linear decay, bf16. As the setup guide asks, we also evaluated the baseline with the chat template and with 5-shot prompts. We did not attempt Traditional Chinese continued pretraining (direction 1) or a new tokenizer or architecture, each of which needs a full 8-hour retraining; the normalizer in 3.1 removes the script handicap at no cost yet brought no gain (Section 4), which made those directions low-priority.

4. Results

ID	Change (from baseline)	Orig. MACRO	Orig. micro	Shuf. MACRO	Shuf. micro	Mean micro	Δ
—	Baseline (MiniMind default)	22.49	24.46	25.21	26.18	25.32	—
—	minimind-3 (author's checkpoint, reference)	23.42	24.89	27.70	26.39	25.64	+0.32
—	Eval option: chat template	23.44	22.53	29.33	26.61	24.57	−0.75
—	Eval option: 5-shot	26.91	27.04	22.98	24.68	25.86	+0.54
T	T→S tokenizer normalizer (no training)	25.89	24.68	22.99	24.25	24.47	−0.85
E1	MC-SFT, no rotation, lr 2e-5, 2 ep	21.58	20.39	22.10	21.24	20.82	−4.50
E2	MC-SFT, rotation, lr 2e-5, 2 ep	31.15	29.61	23.70	24.25	26.93	+1.61
E3	T + MC-SFT, no rotation, lr 2e-5, 2 ep	27.05	25.54	21.05	22.75	24.15	−1.17
E4	T + MC-SFT, rotation, lr 2e-5, 2 ep	29.66	27.90	19.09	21.24	24.57	−0.75
E5	MC-SFT, rotation, lr 1e-4, 3 ep	28.65	27.47	29.37	24.68	26.08	+0.76
E6	MC-SFT, rotation, lr 3e-4, 3 ep	22.80	24.46	26.80	26.39	25.43	+0.11
E7	**MC-SFT, rotation, lr 1e-4, 10 ep (final)**	**33.21**	**28.54**	**31.37**	**27.90**	**28.22**	**+2.90**
R0	Baseline re-trained by run.sh	23.48	24.89	23.80	26.39	25.64	+0.32
R7	E7 recipe re-run by run.sh (on R0)	23.57	24.25	28.72	26.61	25.43	+0.11

Table 2. Held-out (20% of validation set) results (%, 466 questions; micro standard error ≈ 2 points). "Shuf." = same questions with shuffled options. Δ = mean micro minus baseline. Only E7 passes the pre-specified +2-point rule; its full reproduction (R0→R7, Section 5) does not.

On the practice set, the final model (E7) scores 26.24% MACRO (micro 25.59 ± 0.98) against the baseline's 21.91% (23.59 ± 0.95); the reproduced model R7 scores 25.36% (24.89 ± 0.97). Most of this gap reflects the removal of the baseline's letter bias, which both runs achieve; relative to the 25% chance level the gains are only +0.4 to +1.2 MACRO points, within about one standard error.

5. Ablation: which change mattered most

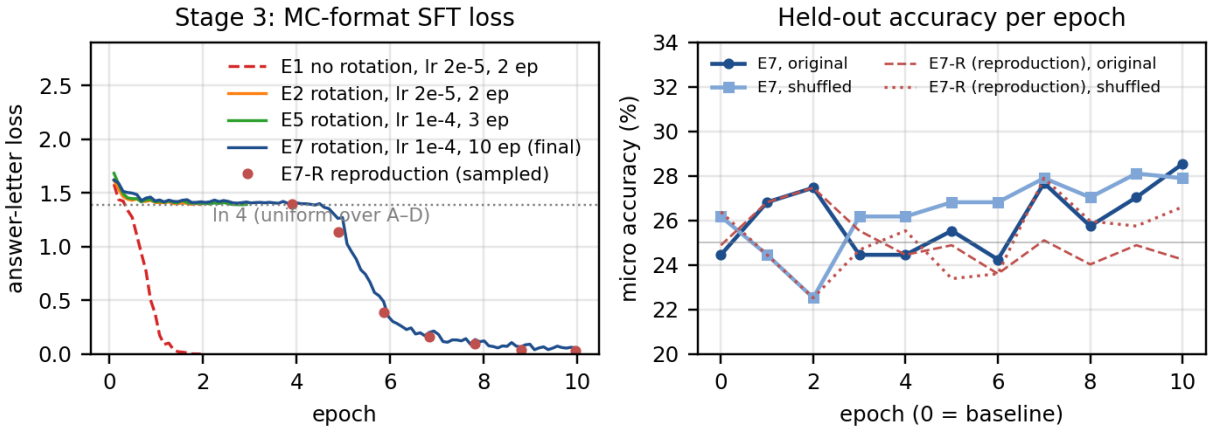


Figure 1. Left: answer-letter loss during Stage 3 (red dots: sampled loss of the reproduction run R7). Right: held-out accuracy after each epoch for E7 and its reproduction R7.

Without rotation the model memorizes: In E1 the loss reaches 0.0015 within two epochs: the model learns a question→letter mapping for the training questions and held-out accuracy drops to 20.4/21.2%, a clear case of overfitting.

With rotation the shortcut disappears, and training length becomes decisive: For two to four epochs, at every learning rate tried (E2, E5, E6), the loss sits at ln(4) ≈ 1.386, the value of a uniform distribution over A–D: the model has learned the format but cannot tell the options apart. Its predictions then appear to collapse onto a single letter: E5 and E6 have identical held-out scores after epochs 1 and 2, and on both held-out versions these scores equal exactly what a model always answering B (epoch 1) or C (epoch 2) would get (those checkpoints were not saved, so this is inferred rather than verified). Only in epoch 5 does the loss break below ln(4), falling to 0.06 by epoch 10: the model learns to locate the correct option wherever it is placed, at least for the training questions. In E7 held-out accuracy then rises in both option orders, but this

did not survive reproduction (below). The T→S normalizer had no consistent effect (E3/E4 vs. E1/E2). Ranking: **rotation is the change that mattered most**, since it is what removes the letter bias and prevents memorization; **training long enough is required for the rotated objective to fit the training set at all**; learning rate and the normalizer had no measurable effect. **No change produced a reproducible gain on unseen questions.**

Is E7's gain real? (i) Its predictions are no longer biased (A 141, B 123, C 87, D 115; Fig. 2). (ii) It beats the score expected from its own letter preference by +3.7 (original) and +2.8 points (shuffled), whereas the baseline beats its expectation by only +0.9 and +0.1. (iii) In a paired comparison with the baseline, the questions answered correctly by only one of the two models split 97 vs. 78 (original) and 94 vs. 86 (shuffled); two-sided sign tests give p = 0.17 and 0.60, so the difference is **not statistically significant** at this sample size. (iv) To rule out leakage we compared every held-out question with all training questions (character 3-gram Jaccard): 5 pairs have similarity ≥ 0.7 (punctuation variants or shared reading passages), and excluding them leaves the results unchanged (28.42/27.77%). The ten epochs were fixed in advance and E7 is the last checkpoint, not one chosen by held-out score.

Reproduction: Re-running the whole pipeline with run.sh reproduces the baseline closely (final losses 1.64/1.70 vs. 1.65/1.71; held-out 24.89/26.39% vs. 24.46/26.18%), as expected given GPU nondeterminism. Stage 3 also behaves the same way: the loss again leaves ln(4) in epoch 5 and ends at 0.03 (Fig. 1), and the epoch-1 and epoch-2 scores are again identical to E7's, which supports the constant-letter interpretation above. **The held-out gain, however, does not reproduce**: R7 scores 24.25/26.61%, at baseline level. E7 and R7 agree on only 46% of held-out predictions, and the difference between them is itself not significant (77 vs. 57 and 68 vs. 62 discordant correct answers, p = 0.10 and 0.66). We conclude that the recipe reliably teaches the answer format, removes the letter bias and fits the training questions, but E7's ~3-point held-out gain most likely reflects run-to-run variance rather than a real ability to answer unseen questions. We kept E7 as the submitted model because it was selected by the pre-specified rule and is statistically indistinguishable from R7.

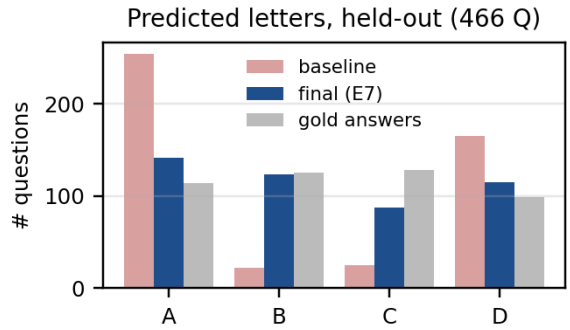


Figure 2. Predicted answer letters on the held-out set (original order) versus the gold distribution.

Micro accuracy (%)	Held-out (orig.)	Held-out (shuf.)	Practice set	Practice MACRO
Baseline	24.46	26.18	23.59	**21.91**
minimind-3 (reference)	24.89	26.39	24.29	**24.73**
Final model (E7)	28.54	27.90	25.59	**26.24**
Reproduction of E7 (R7)	24.25	26.61	24.89	**25.36**

Table 3. Held-out vs. practice set. Baseline, minimind-3 and E7 are ranked identically on all three sets, but the held-out set overstates the size of the gains.

Held-out vs. practice set: Table 3 shows that the held-out set ranks the baseline, minimind-3 and E7 the same way as the practice set, but overstates the gains: E7 leads the baseline by 2–4 points on the held-out set and by 2 points (micro) on the practice set. This is expected, since the held-out set is small (about 7 questions per subject, which makes MACRO especially noisy) and E7 was selected using it. The reproduction R7 illustrates the same point: its held-out scores are 1.3–4.3 points below E7's (2.8 on average), while on the larger practice set the two differ by only 0.7 points (micro). With so few models this comparison is indicative, not conclusive.

6. Training cost

All runs used one **NVIDIA GeForce RTX 4070 Ti SUPER (16 GB)**.

Baseline: pretraining 2.9 h + SFT 5.1 h = **8.0 GPU-hours** (about 7.3 GB of GPU memory at 99% utilization).

Stage-3 experiments: seven runs, 0.50 GPU-hours in total (the final run took 13.4 min).

Evaluations and analyses: under 0.2 GPU-hours. **Development total ≈ 8.7 GPU-hours.**

A full reproduction with run.sh took 8.4 h (pretraining 2.9 h, SFT 5.1 h, Stage 3 0.3 h), for a **grand total of about 17.1 GPU-hours**.

7. Failure cases

On the practice set, all four categories of the final model lie near chance (25.8–26.6%). 30 of 66 subjects are at or below 25%, and six are below 15% (business management 7.1%, education 8.3%, real estate 10.0%, culinary skills 10.3%, TCM clinical medicine 10.7%, physical education 11.1%). However, **subject-level differences are almost entirely sampling noise**: with about 30 questions per subject, the standard error at 25% is about 8 points, close to the observed spread across subjects (SD 9.5 points), and the per-subject accuracies of the baseline and the final model are uncorrelated (Pearson r = 0.006). We therefore cannot claim that any individual subject is reliably above or below chance; no subject is distinguishable from chance.

We attribute this to four causes: **(1) Missing knowledge:** the mini datasets are mostly general-domain Simplified-Chinese text and chat data, while TMMLU+ tests professional, often Taiwan-specific knowledge (law, taxation, dentistry, Taiwanese Hokkien); 64M parameters can store little of it, and the author's minimind-3 checkpoint is also at chance. **(2) Script mismatch:** 38.6% of character occurrences reach the model as unfamiliar byte fragments; the normalizer fixes reading but cannot supply knowledge. **(3) Reasoning and length:** calculation-heavy STEM questions likely require multi-step reasoning beyond a model of this size, and 1.1% of prompts (p99 = 793 tokens) exceed the 768-token SFT length, although this affects too few questions to explain the overall result. **(4) Grounding answers in options:** even on memorized training questions, the model needed five epochs to learn to point to the correct option. Evaluation tricks do not help: the chat template leaves the A/D bias unchanged, and 5-shot prompts merely move the bias to C (231 of 466 predictions).

8. Conclusion and limitations

Format alignment with option rotation, trained long enough for the loss to escape the uniform solution, reliably teaches the answer format and removes the baseline's letter bias, which lifts the practice-set score from 21.9% to about 25–26% MACRO, i.e. back to chance level. It does not reliably make the model answer unseen questions: the held-out gain of our selected run disappeared when the whole pipeline was re-run, and no comparison against the baseline was statistically significant. The T→S tokenizer fix and evaluation options did not help either. Our results suggest that, at 64M parameters trained on general Simplified-Chinese data, the bottleneck on TMMLU+ is knowledge and capacity rather than format or tokenization.

Limitations: the held-out set is small (±2 points); deduplication between splits removed only exact duplicates; about ten configurations were compared, which raises the risk of a lucky winner (as the reproduction suggests); and we ran the final recipe only twice, so a proper estimate of run-to-run variance would need more seeds.

Appendix A. Use of AI assistants

I used Claude (Anthropic) as an assistant throughout HW1:

- **Code:** drafting the scripts convert.py, split_validation.py, analyze_tokenizer.py, add_t2s_normalizer.py, train_mc.py and check_near_duplicates.py, as well as run.sh. I ran all code myself and checked the outputs.
- **Experiment design and analysis:** discussing the evaluation protocol, the pre-specified decision rule, the letter-bias diagnosis, significance testing and the near-duplicate check.
- **Report:** generating the figures from my logs and drafting the text, which I then reviewed and revised. All numbers come from my own runs.

Appendix B. Baseline training curves

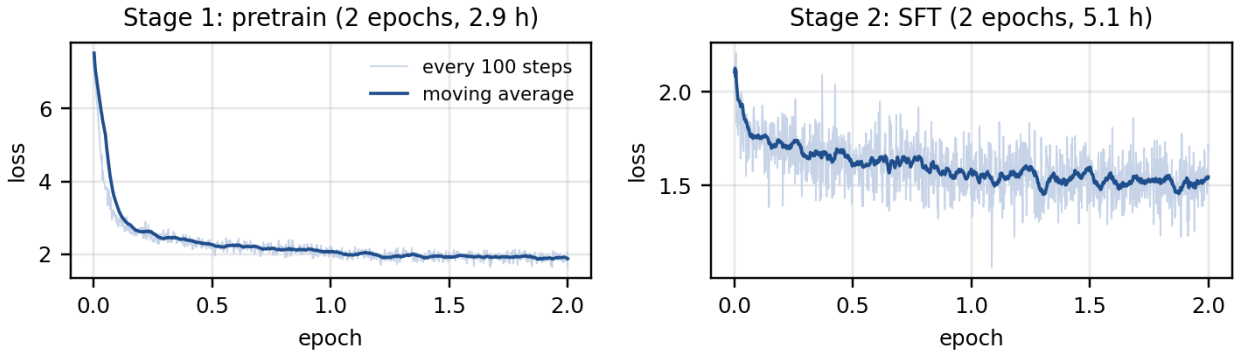


Figure 3. Loss during Stage 1 (pretraining) and Stage 2 (SFT) of the baseline (logs/pretrain.log, logs/sft.log).